

Obfuscation & DPI/ Censorship Landscape

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Volume 2 (Data Plane) nano-detail specification. This document deepens the *Obfuscation & DPI/Censorship Landscape* surface of the pass-1 data-plane overview [01 §3, §3.3–§3.8, §5.3]. It is **SPEC-ONLY**: it describes *what to build* (Rust signatures, wire byte-layouts, state machines, error taxonomy, config knobs, edge cases, security analysis, performance budget, and the §11.4.169 test points), not the shipping product. Every factual claim cites a source id — [01 §N] (the pass-1 data-plane overview), [04_ARCH §N] (HelixVPN-Architecture-Refined), [04_P0 §N] (Phase0-Spike), [SYNTHESIS §N], [research-hysteria2], [research-masque], [research-daita]. Facts not groundable in the evidence base are flagged UNVERIFIED: per §11.4.6.

0. What this document owns (and what it defers)

The pass-1 overview established the layering invariant — *WireGuard is the crypto core; obfuscation/transport is a swappable layer **under** WG, never a fork of WG crypto* [01 §0, §0.1 I1–I6, 04_ARCH §0/§3.1]. It listed the transport set (plain-udp, masque-h3, connect-ip, shadowsocks, udp-over-tcp, lwo, hysteria2) [01 §3] and surfaced decision **D1** (primary obfuscating transport: masque-h3 vs Hysteria2+Salamander) [01 §3.8, SYNTHESIS §3 D1].

This document owns the censorship side of that picture:

#	Owned here	Deferred to
01		—

#	Owned here	Deferred to
	The censorship-regime taxonomy (GFW QUIC-SNI, hard UDP block, SNI/TLS DPI, active probing, throttling) and the formal regime → transport survival matrix	
O2	Passive vs active probing resistance as a typed property of every Transport, plus the edge probe classifier state machine + decoy responder	bin/helix-edge.rs impl (01 §2)
O3	D1 fully analyzed from all angles with the cited tradeoff, and the resolved spec stance (primary + ladder + per-leg)	decision log (04_P0 §7)
O4	The regime-aware escalation ladder that selects the surviving transport per detected regime, refining the flat ladder of [01 §5.3]	helix-core/src/ladder.rs (01 §5.3)
O5	Wire byte-layouts of the obfuscation envelopes (LWO header mangling, Salamander scramble, MASQUE HTTP-Datagram, Phantun fake-TCP) so DPI cannot fingerprint them	the per-transport send/recvd (01 §3)
O6	Anti-fingerprinting (QUIC/TLS ClientHello mimicry) as an explicit open design problem with a typed config seam	security doc
O7	The §11.4.169 test points that prove each survival claim with captured evidence	Volume 8 testing docs

Out of scope (referenced for seams only): the WG crypto core internals [01 §4]; DAITA traffic-shaping internals (its *machines* live in the security doc; only its **placement relative to obfuscation** is noted here at §9) [01 §9, research-daita]; the control-plane push of TransportPolicy [01 §5.3, doc 03]; multi-hop nesting [01 §11.1].

1. The censorship-regime taxonomy

A "censorship regime" is the **set of network-layer capabilities the adversary applies on the path between client and gateway**. HelixVPN does not guess the regime from a label; it **detects it from observed transport failures** (§4) and selects the surviving transport. The closed taxonomy below is the type the detector emits and the survival matrix (§2) is keyed on.

```
// helix-transport/src/regime.rs
//
// A detected (or coordinator-asserted) property of the path. NOT
// a country –
// a capability set. Multiple flags may hold simultaneously (e.g.
// GFW = QUIC-SNI
```

```

// + active-probing + residual-throttle). Bitflags so the survival
// matrix can AND
// the transport's resistance mask against the regime's threat
// mask.
bitflags::bitflags! {
    #[derive(Clone, Copy, Debug, PartialEq, Eq, Default)]
    pub struct Regime: u16 {
        /// Plain WireGuard/UDP to :51820 is dropped or RST – WG
        /// message-type
        /// fingerprint (types 1–4) and/or invariant sizes (init
        /// 148 B,
        /// response 92 B) are matched. [research-hysteria2 §4]
        const WG_FINGERPRINT = 1 << 0;
        /// ALL UDP (incl. QUIC/443) is dropped or throttled to
        /// unusable.
        /// Corporate + China-domestic common. [research-hysteria2
        /// §5(b)]
        const UDP_BLOCK = 1 << 1;
        /// QUIC is permitted BUT the QUIC-Initial SNI is
        /// extracted and matched
        /// against a blocklist (GFW since ~2024-04-07).
        /// [research-hysteria2 §5(a)]
        const QUIC_SNI_FILTER = 1 << 2;
        /// TLS ClientHello SNI on TCP/443 is extracted +
        /// blocklist-matched.
        /// [research-hysteria2 §5(c)]
        const TLS_SNI_FILTER = 1 << 3;
        /// The adversary actively probes the server to confirm a
        /// flagged flow
        /// (Shadowsocks-class). [research-hysteria2 §5(d)]
        const ACTIVE_PROBING = 1 << 4;
        /// Residual/stateful enforcement: after a trigger the 3-
        /// tuple/4-tuple is
        /// blocked for a window (GFW: ~3 min, ~500 ms to engage).
        /// [research-hysteria2 §5(a)]
        const RESIDUAL_BLOCK = 1 << 5;
        /// Bandwidth throttling of the suspected flow (not a hard
        /// drop).
        const THROTTLE = 1 << 6;
        /// Packet-size-distribution flagging feeds
        /// ACTIVE_PROBING. [research-hysteria2 §5(d)]
        const SIZE_PROFILING = 1 << 7;
    }
}

/// The threat a transport must defeat to PASS, expressed as the
/// same flags.
/// A transport `survives` a regime iff `transport.resistance() ≥
/// regime`
/// (every regime threat bit is covered by a resistance bit), with
/// the UDP_BLOCK
/// special-case (§2.2): UDP-carried transports can NEVER cover
/// UDP_BLOCK.
pub type ThreatMask = Regime;

```

1.1 Per-regime mechanism notes (cited facts)

- **QUIC_SNI_FILTER (GFW, USENIX Sec '25).** The GFW derives the QUIC-Initial decryption key from the packet header (RFC 9001 makes the key header-derivable), decrypts the Initial, extracts the SNI, matches a blocklist. It blocked an average of **43.8K FQDNs/week** (Oct 2024-Jan 2025) [research-hysteria2 §5(a)]. **Critical exploitable weakness:** the GFW does **not** reassemble QUIC Initials split across more than one UDP datagram — fragmenting the handshake slips past [research-hysteria2 §5(a)]. This is the design basis for Salamander/Gecko (§5.3).
- **RESIDUAL_BLOCK.** Enforcement is stateful: ~**3-minute** block after a trigger, ~**500 ms** to engage, **58% by 3-tuple / 37% by 4-tuple** [research-hysteria2 §5(a)]. Consequence: a long-lived tunnel is disrupted; a short-lived flow may complete inside the uncensored window. The ladder **MUST** treat a mid-session drop on a previously-working transport as a *residual* event, not a fresh dial failure (§4.4).
- **UDP_BLOCK.** QUIC / Hysteria2 / MASQUE-over-H3 / WireGuard / AmneziaWG **all fail** — they need UDP [research-hysteria2 §5(b)]. Only TCP-carried transports (shadowsocks, udp-over-tcp, MASQUE's HTTP/2 fallback) survive.
- **ACTIVE_PROBING.** The GFW passively flags a suspicious flow (partly by packet-size distribution → SIZE_PROFILING), then actively connects to the server to confirm. Mitigation: a probe-resistant front that returns **nothing useful** to an unauthenticated prober [research-hysteria2 §5(d)] (§3).
- **Mullvad's public stance (the parity target).** Mullvad shipped WG-over-MASQUE QUIC obfuscation 2025-09-09 (desktop v2025.9), auto-tried after failed normal attempts, "resembling standard web activity" — and published **no** performance numbers and **no** explicit undefended threat [research-hysteria2 §2, research-masque §2]. The "looks like HTTP/3" claim rests on **collateral-damage deterrence** (blocking QUIC/443 breaks Google/YouTube/Cloudflare), **not** on a published ClientHello-mimicry layer — that layer is an open problem (§6).

[MERMAID RENDER FAILED — block #1]

2. The regime → transport survival matrix (the load-bearing artifact)

2.1 The matrix (facts cited per cell-source)

Each transport declares a **resistance mask** (ThreatMask); a transport *survives* a regime iff its mask covers every threat bit (subject to the UDP-block carve-out, §2.2). The matrix is the compiled, testable form of [research-hysteria2 §5] and [01 §3.2 transport matrix].

Transport	Carrier	WG_FINGERPRINT	UDP_BLOCK	QUIC_SNI_FILTER	TLS
plain-udp	UDP	✗ (raw WG sig)	✗	n/a (not QUIC)	n/a
lwo	UDP	✓ (header+size mangle)	✗	n/a	n/a
masque-h3	QUIC/ UDP 443	✓ (no WG sig on wire)	✗	✓ iff proxy SNI unblocked/ fronted	n/a
hysteria2+Salamander	QUIC/ UDP	✓	✗	✓ (SNI/Initial fully scrambled)	n/a
hysteria2+Gecko	QUIC/ UDP	✓	✗	✓ (+ handshake fragmentation)	n/a
shadowsocks	TCP	✓	✓	n/a	✓ (ish)
udp-over-tcp	TCP	✓	✓	n/a	✓ i wr
connect-ip	QUIC/ UDP 443	✓	✗	as masque-h3	n/a

Notes that are **facts**, not preferences:

- **masque-h3 vs QUIC_SNI_FILTER is conditional**: it survives *only if* the HTTP/3 proxy's own SNI is unblocked or fronted; otherwise it is SNI-filtered like any QUIC [research-hysteria2 §6]. This is why §6 (anti-fingerprinting/SNI strategy) is load-bearing, not optional.
- **Hysteria2+Salamander hides the SNI/Initial entirely** → survives QUIC_SNI_FILTER unconditionally; **Gecko additionally fragments the handshake** → directly defeats the GFW no-reassembly weakness [research-hysteria2 §1, §5(a)].
- **Neither QUIC path survives UDP_BLOCK alone** — both need a TCP fallback (MASQUE: in-protocol HTTP/2; Hysteria2: pair with Phantun udp2tcp or a Reality/ShadowTLS sibling) [research-hysteria2 §6].
- **AmneziaWG** is the *lowest-overhead* answer to WG_FINGERPRINT-only regimes (~3% throughput cost, still UDP) — HelixVPN realizes this class as the two rung rather than a separate transport [research-hysteria2 §4, 01 §3.7].

2.2 The matrix as code (compile-time + runtime)

```
// helix-transport/src/survival.rs
use crate::regime::{Regime, ThreatMask};
use crate::TransportKind;

/// Each transport's static resistance. Constructed once; unit-
/// tested cell-by-cell
/// against §2.1 (a paired §1.1 mutation flips one cell and a test
/// MUST FAIL).
pub const fn resistance(k: TransportKind) -> ThreatMask {
    use Regime as R;
    match k {
        TransportKind::PlainUdp    => R::empty(),
        TransportKind::Lwo          => R::WG_FINGERPRINT,
        // masque-h3 covers QUIC_SNI_FILTER only when fronted (see
        // `survives`);
        // the static mask claims it, the runtime guard verifies
        // the SNI premise.
        TransportKind::MasqueH3    => R::WG_FINGERPRINT
                                   | R::QUIC_SNI_FILTER |
        R::ACTIVE_PROBING,
        TransportKind::Hysteria2   => R::WG_FINGERPRINT
                                   | R::QUIC_SNI_FILTER |
        R::ACTIVE_PROBING | R::THROTTLE,
        TransportKind::Shadowsocks=> R::WG_FINGERPRINT |
        R::UDP_BLOCK
                                   | R::TLS_SNI_FILTER |
        R::ACTIVE_PROBING,
        TransportKind::UdpOverTcp => R::WG_FINGERPRINT |
        R::UDP_BLOCK,
        TransportKind::ConnectIp  => R::WG_FINGERPRINT
```

```

| R::QUIC_SNI_FILTER |
R::ACTIVE_PROBING,
}
}

/// True iff `k` is expected to carry a tunnel under `regime`.
/// The UDP_BLOCK carve-out is explicit: a UDP-carried transport
    can NEVER survive
/// a hard UDP block, regardless of its other resistance bits
    (§11.4.6 – do not let
/// a mask bug claim the impossible).
pub fn survives(k: TransportKind, regime: Regime) -> bool {
    if regime.contains(Regime::UDP_BLOCK) && carrier_is_udp(k) {
        return false;
    }
    resistance(k).contains(regime & ALL_THREATS_EXCEPT_THROTTLE)
}

pub const fn carrier_is_udp(k: TransportKind) -> bool {
    matches!(k,
        TransportKind::PlainUdp | TransportKind::Lwo
    | TransportKind::MasqueH3 | TransportKind::Hysteria2
    | TransportKind::ConnectIp)
}

const ALL_THREATS_EXCEPT_THROTTLE: Regime =
    Regime::WG_FINGERPRINT.union(Regime::UDP_BLOCK)
    .union(Regime::QUIC_SNI_FILTER).union(Regime::TLS_SNI_FILTER)
    .union(Regime::ACTIVE_PROBING).union(Regime::RESIDUAL_BLOCK)
    .union(Regime::SIZE_PROFILING);

```

Edge case — the conditional masque-h3 cell. `resistance()` statically claims `QUIC_SNI_FILTER` for `masque-h3`, but that claim holds *only* when the proxy SNI is fronted/unblocked. The dial path (§6.2) carries an `sni_strategy`; if the strategy is `PlainBlockable` and the regime has `QUIC_SNI_FILTER`, the ladder **MUST** treat `masque-h3` as **not surviving** and skip to `hysteria2/shadowsocks`. This is a runtime refinement of the static mask, asserted by a dedicated test (§7 T-D1-c).

3. Passive vs active probing resistance (typed, edge-enforced)

3.1 The two probing models

- **Passive probing** = the adversary only observes. Resistance = *the bytes on the wire carry no transport-distinctive signature*: no WG message-type, no invariant size, no plaintext SNI, no

QUIC-Initial structure. This is a property of the **client-side envelope** (§5).

- **Active probing** = the adversary, having flagged a flow, *connects to the gateway* and replays/forges traffic to elicit a distinctive response. Resistance = *the edge returns nothing useful to an unauthenticated peer* — it must be **indistinguishable from a benign server** to a prober [research-hysteria2 §5(d)].

Both are required for a QUIC_SNI_FILTER + ACTIVE_PROBING regime (the GFW). The client envelope defeats passive; the edge probe-classifier defeats active.

```
// helix-transport/src/probe.rs
/// A transport declares, as a type, what kind of probing it
/// defeats.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub struct ProbeResistance {
    /// Passive: no transport-distinctive bytes observable on the
    /// wire.
    pub passive_unfingerprintable: bool,
    /// Active: an unauthenticated prober receives an
    /// indistinguishable-from-benign
    /// response (decoy site / silent drop / RST as a real closed
    /// port would).
    pub active_decoy_or_silent: bool,
    /// The auth secret that gates a "useful" response (separate
    /// from WG keys).
    pub auth_gate: AuthGate,
}
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
pub enum AuthGate {
    /// Salamander/LW0 PSK: wrong PSK -> packet is
    /// indistinguishable noise, edge
    /// silently drops (no response at all). [research-hysteria2
    /// §1, §5(d)]
    PreSharedKey,
    /// Hysteria2 password over the in-QUIC HTTP/3 auth exchange:
    /// wrong password ->
    /// the H3 server returns a generic 404/decoy, never a
    /// protocol tell. [research-hysteria2 §1]
    Hysteria2Password,
    /// masque-h3: a non-CONNECT-UDP request is served the decoy
    /// site (§3.3).
    MasqueExtendedConnect,
    /// Shadowsocks AEAD: a forged/replayed first frame fails AEAD
    /// -> edge keeps the
    /// TCP open and behaves like a stuck/slow benign server (no
    /// RST tell). [research-hysteria2 §5(d)]
    AeadFirstFrame,
}
```


3.2 The edge probe-classifier state machine (bin/helix-edge.rs)

The :443 (and the regime-fallback :443/tcp) listener classifies every new flow **before** committing transport resources. Native edge behavior replacing the original Nginx-camouflage [01 §3.3, 04_ARCH §3.3].

[MERMAID RENDER FAILED — block #2]

```
// bin/helix-edge.rs (classifier core – spec sketch)
pub enum FlowClass {
    Tunnel(TransportKind), // authenticated -> hand to the
                           // transport de-obfuscator
    Decoy,                 // serve the believable decoy site
                           // (probe sees a real website)
    SilentDrop,            // no response at all (Salamander/PSK
                           // miss)

    BenignRst,             // TCP: mimic a closed/benign
                           // port, no protocol tell
}

/// MUST be constant-time w.r.t. the auth decision to avoid a
/// timing oracle
/// distinguishing "wrong PSK" from "no PSK" (§3.4 security). MUST
/// NOT log the
/// peer addr durably (I5, no per-connection state).
pub fn classify(first: &FirstBytes, decoy: &DecoySite) ->
    FlowClass { /* ... */ }
```

3.3 The decoy responder (masquerade)

For any flow that is **not** a valid tunnel, the :443 listener serves a believable decoy site so a prober/scanner sees an ordinary HTTP/3 (or HTTPS) server [01 §3.3, research-masque §2]. Requirements:

- The decoy MUST be a *complete* small site (valid TLS cert for the gateway's hostname, real HTML, 200 OK on /), not an empty 444/blank — a blank response is itself a fingerprint.
- The decoy hostname's certificate and the tunnel's SNI MUST be the **same** name so SNI inspection of a probe vs a real client cannot distinguish them.
- The decoy MUST be served by the same QUIC/TLS stack (same ClientHello-response fingerprint) as the tunnel path, so a probe cannot tell "tunnel server" from "web server" by transport fingerprint.

3.4 Security analysis of probing resistance

Threat	Defense	Residual risk
Timing oracle (auth-fail vs no-auth latency)	constant-time <code>classify()</code> ; identical code path to first I/O	UNVERIFIED: exact constant-time guarantee of the underlying QUIC lib — must be measured (§7 T-SEC-a)
Replay of a captured authenticated first frame	Shadowsocks AEAD nonce / Hysteria2 session binding rejects replay; edge keeps TCP open (BenignRst path) so replay yields no tell	replay within a live AEAD window — bounded by per-session key
Decoy/tunnel TLS fingerprint mismatch	same QUIC/TLS stack serves both	the <i>client</i> ClientHello mimicry is the harder open problem (§6)
Size-distribution profiling (SIZE_PROFILING) feeding active probing	LWO/Salamander randomized padding flattens the size histogram (§5); DAITA (§9) is the strong form	padding is heuristic; DAITA is the calibrated answer (security doc)
Durable per-connection state revealing who connected	I5 no-logging by construction; classifier holds only in-flight state	none by construction [01 §0.1 I5]

4. The regime-aware escalation ladder (refines [01 §5.3])

4.1 What changes vs the flat ladder

[01 §5.3] walks a static `TransportPolicy.order` on repeated handshake failure. This document refines that into a **detector-driven** ladder: each failure updates a Regime estimate, and the next rung is chosen as *the highest-throughput transport whose survives(k , regime) is true* — not merely the next array element. The flat order remains the fallback when detection is inconclusive.

```
// helix-core/src/ladder.rs (extends the §5.3 sketch)
pub struct RegimeAwareLadder {
    order: Vec<TransportKind>,          // coordinator-pushed prior
                                       // (§5.3 step 5)
    pin: Option<TransportKind>,         // manual override (skip
                                       // detection)
    budget: FailureBudget,              // { max_handshakes,
                                       // window } per rung
    estimate: Regime,                  // accumulated from failure
                                       // signatures (§4.3)
    per_network_memory: PerNetworkMap // SSID/gw-fingerprint ->
                                       // last working kind (§5.3 step 4)
}

impl RegimeAwareLadder {
    /// Choose the next rung. Preference: survives(estimate) AND
    /// highest throughput
    /// class, breaking ties by `order` position (coordinator
    /// prior).
    pub fn next_rung(&self) -> Option<TransportKind> { /* ... */ }

    /// Fold a dial/runtime failure into the regime estimate (§4.3
    /// mapping).
    pub fn observe(&mut self, k: TransportKind, err:
        &TransportError, phase: FailPhase);
}

#[derive(Clone, Copy, Debug)]
pub enum FailPhase { Dial, Handshake, MidSession }
```

4.2 Failure-signature → regime inference (the detector)

The detector turns concrete `TransportErrors` (the taxonomy of [01 §3.1 error.rs]) plus the *phase* into `Regime` bits. This is the only place a regime is *inferred*; it is heuristic and **MUST** be honest about uncertainty (§11.4.6 — an inference is a hypothesis until a working transport confirms it).

Observed	Phase	Inferred bit(s)	Rationale (cited)
plain-udp → HandshakeFailed but two overmangled UDP → Connected	Handshake	WG_FINGERPRINT	only the WG signature was caught [research-hysteria2 §4]
every UDP transport → DialTimeout/HandshakeFailed; a	Dial/HS	UDP_BLOCK	UDP path dead, TCP alive [research-

Observed	Phase	Inferred bit(s)	Rationale (cited)
TCP transport → Connected			hysteria2 §5(b)]
masque-h3 plain SNI → EndpointBlocked; fronted SNI → Connected	Handshake	QUIC_SNI_FILTER	SNI was the discriminator [research-hysteria2 §5(a)]
Connected then MidSession drop ~minutes in, redial blocked ~3 min then works	MidSession	RESIDUAL_BLOCK	stateful 3-tuple/4-tuple block window [research-hysteria2 §5(a)]
Connected but goodput collapses while ping survives	MidSession	THROTTLE	throttle not drop
handshake completes but server appears probed (out-of-band signal, coordinator-pushed)	—	ACTIVE_PROBING	passive flag → active probe [research-hysteria2 §5(d)]

```
// helix-core/src/ladder.rs
fn infer(err: &TransportError, phase: FailPhase, last_ok:
    Option<TransportKind>) -> Regime {
    use TransportError as E; use FailPhase as P;
    match (err, phase) {
        (E::HandshakeFailed(_), P::Handshake) if last_ok ==
            Some(TransportKind::Lwo)
            => Regime::WG_FINGERPRINT,
        (E::DialTimeout, P::Dial) | (E::HandshakeFailed(_),
            P::Handshake)
            => Regime::empty(), // ambiguous – caller ANDs across
            multiple observations
        (E::EndpointBlocked, _) => Regime::QUIC_SNI_FILTER |
            Regime::TLS_SNI_FILTER, // narrowed by carrier
        (E::Closed, P::MidSession) => Regime::RESIDUAL_BLOCK,
        _ => Regime::empty(),
    }
}
```

Honest boundary (§11.4.6): a single DialTimeout is ambiguous (could be UDP block, could be packet loss). The detector only *commits* a bit after the **contrast** observation (the paired success of a differently-carried transport) or a coordinator-pushed regional prior. Until then estimate stays conservative and the

ladder falls back to order. The success of a transport is the captured evidence that confirms the regime (the per-network memory then short-circuits next time — [01 §5.3 step 4]).

4.3 Regime-aware escalation sequence

[MERMAID RENDER FAILED — block #3]

4.4 Residual-block handling (the GFW-specific edge case)

When MidSession drop is inferred as RESIDUAL_BLOCK, the ladder **MUST not** immediately re-dial the same transport on the same 3-tuple (it is blocked for the window). Required behavior:

1. Mark the (transport, local-port) tuple cooling for the configured residual_cooldown (default 200 s, > the ~180 s observed GFW window [research-hysteria2 §5(a)]; UNVERIFIED: the exact per-PoP window — make it a config knob, §6.3).
 2. Re-dial the *same logical transport* from a **fresh local port** (changes the 4-tuple) OR escalate to the next surviving rung — whichever the policy prefers (residual_strategy: { RotatePort | Escalate }).
 3. Emit TunnelStatus::Reconnecting with a residual reason tag so the UX/telemetry distinguishes censorship-residual from ordinary roaming.
-

5. Wire byte-layouts of the obfuscation envelopes (so DPI cannot match)

This section specifies the **exact bytes on the wire** for each obfuscation layer, because "looks like X" is only testable against a byte spec. WG's own datagram is the opaque payload in every case (I1).

5.1 The WireGuard signature DPI keys on (what we must hide)

WireGuard transport/handshake datagram (the thing DPI fingerprints):

```
offset 0      1      2      3      4 .....
      +-----+-----+-----+-----+-----+
      | type |      reserved (3)      | (type-specific body)      |
      +-----+-----+-----+-----+-----+
type ∈ {1=handshake_init, 2=handshake_resp, 3=cookie,
4=transport_data}
```

INVARIANT SIZES the GFW/DPI match: init = 148 bytes, response =

92 bytes.

[research-hysteria2 §4]

DPI matches on: (a) the fixed 1-byte type in {1..4} at offset 0, (b) the 3 reserved zero bytes, (c) the invariant 148/92 sizes of handshakes. An obfuscation layer must break **all three** to defeat WG_FINGERPRINT [research-hysteria2 §4].

5.2 lwo — lightweight WG header obfuscation (Phase 1 basic → Phase 2 keyed)

lwo on-wire datagram (per session key K, 8-byte epoch nonce N):

```
+-----+-----+
+-----+
| N (8) | mangled WG datagram | random padding
(0..P) |
+-----+-----+
+-----+
mangle = bitwise XOR of the FIRST 4 bytes (type+reserved) with
        keystream = HKDF(K, N)[0..4] ; body bytes unchanged
(cheap)
P (pad length) drawn per-packet from a per-session-seeded RNG so
the size
```

histogram no longer shows the invariant 148/92 spikes (defeats size match).

Phase 1: XOR only the 4 header bytes + random pad (basic). [01 §3.7]

Phase 2: keyed per-session scheme over a wider prefix + size-distribution shaping.

```
// helix-transport/src/lwo.rs (envelope only; WG payload is
    opaque)
pub struct Lwo { sess_key: SecretBytes, epoch: [u8;8], pad_rng:
    PadRng, /* sock */ }
impl Lwo {
    fn wrap(&self, wg: &[u8], out: &mut Vec<u8>) { // send
        path
        out.clear();
        out.extend_from_slice(&self.epoch); // N
        let ks = hkdf4(&self.sess_key,
            &self.epoch); // 4-byte keystream
        out.extend_from_slice(wg);
        for i in 0..4 { out[8+i] ^=
            ks[i]; } // mangle type+reserved
        let p =
            self.pad_rng.next_len(); // 0..=P, defeats
            size match
        out.resize(out.len()+p as usize, 0);
        self.pad_rng.fill(&mut out[out.len()-p as usize..]);
    }
}
```

```

    fn unwrap(&self, dg: &[u8]) -> Result<Bytes, TransportError>
        { /* strip N, de-XOR, drop pad */ }
}

```

Edge cases: (1) the padding MUST be cryptographically-seeded random, never a constant byte (a constant tail is itself a signature); (2) wrap MUST reject a WG datagram + max pad that would exceed `effective_mtu()` (`TransportError::Oversize`) [01 §3.1, §10]; (3) the epoch `N` rotates per the WG rekey so a captured keystream ages out.

5.3 hysteria2 Salamander / Gecko — full QUIC scramble (feature-gated)

Salamander obfuscation of EVERY QUIC packet [research-hysteria2 §1]:

```

+-----+
+-----+
| salt(8) | scrambled = QUIC_packet XOR
keystream |
+-----+
+-----+
keystream = BLAKE2b-256( PSK || salt )      (salt = random 8
bytes/packet)
Result: no QUIC-Initial structure, no SNI, no recognizable QUIC
header on wire.
-> defeats QUIC_SNI_FILTER (nothing to extract) AND
WG_FINGERPRINT (WG is inside QUIC).

```

Gecko (experimental, builds on Salamander) [research-hysteria2 §1]:
 additionally FRAGMENTS the QUIC handshake into randomly-sized,
 randomly-padded
 chunks (`config min_packet_size / max_packet_size`) -> defeats the
 GFW's
 single-datagram QUIC-Initial inspection (no reassembly).
 [research-hysteria2 §5(a)]

```

// helix-transport/src/hysteria2.rs (feature = "hysteria2")
#[cfg(feature = "hysteria2")]
pub struct Salamander { psk: SecretBytes } // 32-byte+ pre-
    shared key
#[cfg(feature = "hysteria2")]
impl Salamander {
    fn scramble(&self, quic_pkt: &mut Vec<u8>, salt: [u8;8]) {
        let ks = blake2b256(&[&self.psk[..], &salt[..]]); // 32-
        byte keystream block
        // XOR the whole packet, keystream re-keyed per 32-byte
        block by counter
        xor_keystream(quic_pkt, &self.psk, &salt);
        // wire = salt(8) || scrambled
    }
}

```

```
}
#[cfg(all(feature = "hysteria2", feature = "gecko"))]
pub struct Gecko { inner: Salamander, min_packet_size: u16,
    max_packet_size: u16 }
```

Facts that bound this: apernet/hysteria latest is **app/v2.9.2 (2026-05-23)**, which *adds Gecko* and *fixes a UDP ACL-bypass security issue* — pin $\geq$ this version and track the advisory [research-hysteria2 §1]. Salamander gives two independent secrets (PSK for wire obfuscation + password for authz) [research-hysteria2 §1] — both wired to the AuthGate of §3.1. Congestion: Hysteria2's **Brutal** ignores loss ("bandwidth cheating"), the reason it wins on throttled/lossy links — but it is *aggressive* and **MUST** be rate-capped to avoid harming the gateway uplink (§6.3 knob) [research-hysteria2 §1, §6].

5.4 masque-h3 — HTTP-Datagram over QUIC (the RFC-standard envelope)

Per WireGuard datagram, MASQUE envelope [research-masque §1, 01 §3.3]:

```
WG datagram (opaque)
└─ HTTP Datagram (RFC 9297): Quarter-Stream-ID (varint) ||
payload
└─ QUIC DATAGRAM frame (RFC 9221, unreliable; type
0x30/0x31)
└─ QUIC short-header packet to https://gateway:443
(HTTP/3)
```

Control plane, established ONCE at dial() [research-masque §1]:
Extended CONNECT (RFC 9220 :protocol = connect-udp)
:method = CONNECT
:scheme = https
:authority = gateway-host
:path = /.well-known/masque/udp/{target_host}/{target_port}/
-> proxy opens UDP socket to the WG endpoint, maps QUIC
DATAGRAM <-> UDP.
Thereafter: NO per-packet HTTP round trip; WG rides as HTTP
Datagrams. [01 §3.3]

```
// helix-transport/src/masque.rs (framing helpers – hand-rolled
    per research-masque §3)
/// RFC 9297 framing: prepend the Quarter-Stream-ID varint (ctx 0
    here) then the WG bytes.
pub fn encode_http_datagram(qsid: u64, wg: &[u8]) -> Bytes { /*
    varint(qsid) || wg */ }
pub fn decode_http_datagram(dg: Bytes) -> Result<Bytes,
    TransportError> { /* strip varint */ }
```

Maturity fact (drives D5/D1, §6.4): there is **no turnkey Rust crate** for RFC 9298 CONNECT-UDP client+proxy — it is hand-rolled on quinn (+ h3 / h3-datagram v0.0.2, Extended CONNECT)

exactly as Mullvad did (quinn_udp evidence) [research-masque §3]. Go has quic-go/masque-go v0.3.0 (2025-06-24), turnkey client+proxy [research-masque §4]. The **HTTP/2 fallback** of RFC 9298/9297 is the in-protocol answer to UDP_BLOCK (degraded, TCP-over-TCP-ish, but connects) [research-hysteria2 §2, research-masque §1].

5.5 TCP-carried fallbacks for UDP_BLOCK

shadowsocks (TCP, AEAD) [01 §3.5]:

per WG datagram: [len(2, big-endian) | WG datagram] -> AEAD-encrypt (chacha20-poly1305

or aes-256-gcm) -> TCP. AEAD tag + length give the probe-resistant first frame (§3.1 AeadFirstFrame).

effective_mtu ~1380 (TCP MSS - 2-byte len - AEAD tag) [01 §10].

udp-over-tcp (last resort) [01 §3.6]:

[len(2) | WG datagram] over ONE TCP connection (Mullvad udp2tcp parity); optional

TLS wrap (tls_sni) to look like HTTPS. Accept head-of-line blocking. effective_mtu ~1380.

Phantun-class udp2tcp (reference, research-hysteria2 §5(b)):

masquerades UDP as a FAKE TCP stream through L3/L4 NAT/firewall, 12-byte overhead,

user-mode TCP state machine (no real retransmit penalty).

Documented pairing for

UDP-blocked nets; HelixVPN's udp-over-tcp is the in-tree equivalent rung.

6. Decision D1 — primary obfuscating transport, fully analyzed

DECISION D1 — primary obfuscating transport.

SURFACED, not silently resolved [SYNTHESIS §3 D1, 01 §3.8, 04_ARCH §0].

6.1 The two camps (cited, not editorialized)

- **Camp A — masque-h3 primary** (CLD/04_ARCH, [01 §3.8]). WG-over-MASQUE (RFC 9298/9297/9221) is **literal Mullvad parity** — "Mullvad's QUIC mode IS WG-over-MASQUE" [04_ARCH §0]. One Rust transport crate shared client↔edge (I4). WG stays the crypto core unchanged.
- **Camp B — Hysteria2 + Salamander/Gecko primary** (plurality of the 10-LLM analyses, [SYNTHESIS §3 D1]). Mature, turnkey QUIC-obfs stack; **best throughput on throttled/lossy links** (Brutal); **out-of-the-box GFW-QUIC**

evasion via Gecko fragmentation; far more mature self-host tooling — but a **non-standard protocol** whose obfuscation is *random noise* rather than *genuine web traffic* [research-hysteria2 §6].

6.2 The tradeoff matrix (verbatim axes from the research, cited)

Axis	Hysteria2- primary (+Salamander/ Gecko)	MASQUE- primary (RFC 9298)	Source
Standardisation	custom protocol, single main impl (apernet) + sing-box	IETF RFC 9298 , multiple impls (masque-go, Cloudflare, Apple Private Relay)	[research-hysteria2 §6]
Throughput on lossy/throttled	best — Brutal ignores loss ("bandwidth cheating")	standard QUIC CC (fair, lower on lossy)	[research-hysteria2 §6]
Looks like normal web	Salamander = random noise (hides QUIC, <i>is</i> unusual UDP); Gecko fragments	strong — genuinely <i>is</i> HTTP/3 to an HTTP proxy; blends with web	[research-hysteria2 §6]
GFW QUIC-SNI (2024-25)	survives via Salamander (no SNI) / Gecko (frag)	survives iff proxy SNI unblocked/ fronted	[research-hysteria2 §6]
Hard UDP block	fails (needs UDP) — add Phantun	in-protocol HTTP/2 fallback	[research-hysteria2 §6]
Self-host maturity	very mature, one-click installers, low-RAM	newer; masque-go turnkey, Rust hand-rolled	[research-hysteria2 §6, research-masque §3/§4]
WG compatibility	tunnels arbitrary UDP incl. WG	designed to carry WG UDP (Mullvad's exact model)	[research-hysteria2 §6]

Axis	Hysteria2- primary (+Salamander/ Gecko)	MASQUE- primary (RFC 9298)	Source
Throughput cost (absolute)	lower (single QUIC layer + WG inside)	higher — double crypto + double CC (WG-in-QUIC), Mullvad- acknowledged	[research- masque §2]

6.3 The resolved spec stance (the recommendation, with the seam)

Adopt Camp A (masque-h3) as the primary obfuscating transport for MVP [01 §3.8], because: (1) it preserves the *WG-crypto-core, pluggable-transport* invariant — a Hysteria2-primary design makes Hysteria2 *the* protocol, weakening I1/I4 [01 §3.8]; (2) it is the only literal Mullvad-parity path [research-hysteria2 §2, research-masque §2]; (3) it keeps one Rust transport crate shared with the edge (I4) [01 §3.8, D5].

Provide hysteria2 (Salamander + optional Gecko) as an additional Transport impl behind a Cargo feature [01 §3.8], because it is strictly better on the two axes MASQUE loses (throttled-link throughput via Brutal; out-of-the-box GFW-QUIC evasion via Gecko). It serves three concrete roles:

1. **A ladder rung** — the surviving transport when THROTTLE or a still-effective QUIC_SNI_FILTER defeats fronted MASQUE (§2.2 / §4).
2. **A Phase-0 unblock** — if hand-rolled masque-h3-in-Rust proves a quagmire, prototype MASQUE first in Go (masque-go) to clear gate G2, then port; meanwhile hysteria2 gives a turnkey QUIC-obfs path [01 §3.8, 04_P0 §13, research-masque §4].
3. **An asymmetric-per-leg option** (decision D6) — Hysteria2 on the user↔gateway leg only, plain WG on the gateway↔connector leg [01 §11.3, 11_MST].

The config knobs the seam exposes (defaults are spec-fixed, all tunable per §11.4.35):

```
// helix-transport/src/config_knobs.rs
pub struct ObfsKnobs {
    pub sni_strategy: SniStrategy,           // §6.4 – fronted vs
        plain (default Fronted)
    pub masque_congestion: Congestion,      // Cubic default; Bbr for lossy mobile
        [01 §3.3]
    pub hysteria2_brutal_mbps: Option<u32>, // cap Brutal send
        rate to protect uplink (REQUIRED if enabled)
```

```

pub gecko_min_packet_size:
    u16, // default 100 (frag chunk floor)
    [research-hysteria2 $1]
pub gecko_max_packet_size: u16, // default 1200 (<
    IPv6_min MTU) [research-hysteria2 $1]
pub residual_cooldown_s: u32, // default 200 (>
    ~180s GFW window) [research-hysteria2 $5(a)]
pub residual_strategy: ResidualStrategy, // RotatePort |
    Escalate (default RotatePort)
pub lwo_max_pad: u16, // default 64 -
    size-histogram flattening
}
#[derive(Clone, Copy, Debug)] pub enum SniStrategy { Fronted,
    PlainBlockable, DomainFront }
#[derive(Clone, Copy, Debug)] pub enum ResidualStrategy {
    RotatePort, Escalate }

```

6.4 The anti-fingerprinting / SNI open problem (O6 — do NOT mark solved)

The hardest, **publicly-unsolved** part of MASQUE parity is making the client's QUIC-Initial / TLS ClientHello *indistinguishable from a real browser* — ALPN h3, realistic SNI, Chrome/Firefox-like transport params [research-masque §2, §5]. Facts:

- Mullvad publishes **no** ClientHello-mimicry/uTLS layer, SNI value, ALPN, or domain-fronting detail — its "looks like HTTP/3" rests on collateral-damage deterrence, not on a documented mimicry layer [research-masque §2].
- Go's masque-go gives **weaker** fingerprint control than Rust+quinn (harder to do uTLS-equivalent in Go) — a point *for* the Rust edge (D5) precisely because fingerprint control is where DPI resistance hinges [research-masque §4, §5].

Spec stance: treat QUIC/TLS fingerprint mimicry as an **explicit open design / risk item**, not a checkbox. The SniStrategy enum (§6.3) is the seam: Fronted (proxy behind an unblocked CDN SNI), DomainFront (TLS SNI ≠ HTTP :authority where the fronting CDN permits), PlainBlockable (no mimicry — the honest baseline that the survival matrix down-grades under QUIC_SNI_FILTER, §2.2). The chosen strategy and its captured wire evidence (a tshark capture classified as HTTP/3 with **no WG signature**, the G2 check [01 §3.3, 04_P0 §8]) are the proof; "it looks like a browser" without that capture is a §11.4.6 guess.

[MERMAID RENDER FAILED — block #4]

7. Test points (§11.4.169) — proving each survival claim with captured evidence

Every survival claim in §2 and every detector inference in §4 is a falsifiable test. The rig is the [research-daita §5] Linux-netns + nftables-DPI-sim + tc netem stack, booted on-demand via the containers submodule (§11.4.76). Each leaf records an evidence path in test_diary (§11.4.149); no metadata-only PASS.

Test id	§11.4.169 types	What it proves	Rig / evidence	Sc
T-MATRIX	UNIT	resistance()/survives() match §2.1 cell-by-cell	table-driven Rust unit test; paired §1.1 mutation flips one cell → FAIL	§2
T-WGFP	INT,E2E	plain-udp blocked by a WG-type nftables match, two survives	netns; nft rule matching type∈{1..4}+148/92 sizes; tshark shows mangled header	[re hy §4 re da
T-UDPBLOCK	INT,E2E	under nft drop udp, shadowsocks/udp-over-tcp connect, all UDP transports fail	netns drop-all-UDP; tunnel up over TCP; iperf3 goodput	[re hy §5 re da
T-QUIC-SNI	INT,E2E,SEC	plain-SNI masque-h3 blocked, fronted-SNI/hysteria2+Salamander survive	netns DPI sim extracting QUIC-Initial SNI (mimics GFW key-derivation); tshark no-SNI on Salamander	[re hy §5
T-GECKO-FRAG	INT	Gecko handshake fragmentation slips a single-datagram-Initial inspector	DPI sim that only inspects 1st datagram; capture multi-datagram Initial	[re hy §5
T-PROBE	SEC,FA	active prober to :443 gets decoy/silent/RST, never a tunnel tell	scripted prober (forged first bytes); assert decoy HTML / no-response / benign-RST; re-runnable -count=3	§3 [re hy §5
T-SEC-a	SEC	classify() auth decision is constant-time (no timing oracle)	timing histogram wrong-PSK vs no-PSK; assert overlap	§3 (UN un lib mu me
	INT,CHAOS			

Test id	§11.4.169 types	What it proves	Rig / evidence	Sc
T-RESIDUAL		mid-session RESIDUAL_BLOCK triggers cooldown + port-rotate, not blind redial	netns stateful-block qdisc; assert Reconnecting{residual} + fresh 4-tuple after window	\$4 [re hy \$5
T-LADDER	FA,E2E	detector commits the right Regime and survives() picks the surviving rung	full ladder run; assert ordered TunnelStatus trace; -count=3 deterministic (§11.4.50)	\$4
T-MTU	UNIT,PERF	each envelope's effective_mtu() is measured, 0oversize enforced	wrap a max datagram; assert TransportError::0versize; measure on wire	[0 \$5
T-G2-WIRE	REC,SEC	masque-h3 flow classifies as HTTP/3 with no WG signature	tshark capture (the G2 wire-fingerprint check) + window-scoped MP4 vision verdict (§11.4.159)	[0 04
T-THROTTLE	PERF,STRESS	Hysteria2/Brutal sustains goodput under netem throttle where MASQUE collapses	tc netem rate+loss; iperf3 goodput bars; assert Brutal cap honored	[re hy \$6 re da \$5
T-LOSS	PERF	under netem loss 5%, masque-h3/QUIC beats the udp-over-tcp strawman	Gilbert-Elliott bursty loss; goodput compare	[0 re da

Anti-bluff floor (§11.4 / §11.4.107): a survival claim is complete only when its T-id is green **with** a captured artifact — a tshark capture, an iperf3 CSV, a probe-response transcript — recorded in test_diary; the wire-fingerprint claims additionally carry a window-scoped MP4 + media-validation verdict (§11.4.159/163). "The transport survives regime X" with no captured DPI-sim evidence is a §11.4 PASS-bluff.

8. Performance budget (the honest cost of obfuscation)

Transport	Relative throughput vs plain-udp	CPU cost	Why	Source
plain-udp		lowest	no obfuscation	

Transport	Relative throughput vs plain-udp	CPU cost	Why	Source
	1.00 (the G1 $\geq 80\%$ -bare-link baseline)			[01 §3.2, 04_P0 §8]
lwo	~0.98 (–padding)	near-zero	XOR 4 header bytes + pad	[01 §3.7]
masque-h3	< 1.0, materially	highest	double crypto + double congestion control (WG-in-QUIC, two AEAD layers) — Mullvad-acknowledged "computationally very expensive"	[research-masque §2]
hysteria2+Salamander	higher than MASQUE on lossy/throttled	moderate	single QUIC layer; Brutal ignores loss	[research-hysteria2 §6]
shadowsocks/udp-over-tcp	moderate (HoL penalty)	moderate	TCP HoL blocking, AEAD	[01 §3.5/ §3.6]

Spec requirement (§11.4.6, anti-bluff): the MASQUE throughput penalty **MUST** be **measured** (WG-direct vs WG-over-MASQUE, the Phase-0 G2 record: MTU / throughput / CPU vs plain-udp) — do **not** claim parity speed [research-masque §5, 01 §10, 04_P0 §5.3]. The numeric penalty is UNVERIFIED: until the G2 benchmark lands; Mullvad published no figures [research-hysteria2 §2]. MTU budget per envelope is fixed at [01 §10] (plain-udp 1420, masque-h3 1280-measure-and-tune, shadowsocks/udp-over-tcp ~1380, lwo ~1400).

9. Relationship to DAITA (placement only — internals deferred)

DAITA (size/timing/frequency defense) sits at **L2.5 — above WG, below the transport** [01 §9, research-daita §2]. Its relationship to *this* document's concern: DAITA is the **calibrated** answer to SIZE_PROFILING (which feeds ACTIVE_PROBING), where the LWO/Salamander per-packet random padding (§5) is the **cheap**

heuristic form. Both flatten the size histogram; DAITA additionally normalizes *timing* and injects bidirectional cover traffic [research-daita §2].

Ordering invariant (load-bearing): DAITA shapes the **stream of WG datagrams before** they hit the obfuscating transport — so the transport obfuscates an *already-shaped* stream. The obfuscation envelope (§5) **MUST NOT** undo DAITA's size normalization (e.g. lwo padding is *additive* to DAITA's constant-size padding, not a replacement). DAITA's machines (maybe not 2.2.2 core, server-pushed v2 model) are config from the NetworkMap, specified in the security doc [research-daita §1/§4, 01 §9]; this document only fixes the placement and the no-conflict rule.

10. Edge cases & failure-mode catalogue (implementation-ready)

#	Edge case	Required behavior	Cite
EC1	Ambiguous single DialTimeout	do NOT commit a regime bit; fall back to order; commit only on contrast success	§4.2
EC2	masque-h3 with PlainBlockable SNI under QUIC_SNI_FILTER	treat as not-surviving; skip to hysteria2/shadowsocks	§2.2, §6.4
EC3	GFW residual block mid-session	cooldown (default 200 s) + port-rotate or escalate; emit Reconnecting{residual}	§4.4
EC4	LWO/Salamander padding pushes datagram > MTU	TransportError::Oversize; orchestrator lowers inner WG MTU	§5.2, [01 §10]
EC5	Hysteria2 Brutal uncapped harms gateway uplink	hysteria2_brutal_mbps is REQUIRED when the feature is enabled; reject config if unset	§6.3
EC6	Active prober replays a captured AEAD/auth first frame	nonce/session binding rejects; edge stays on BenignRst path, no tell	§3.4
EC7	Decoy site returns blank/444	forbidden — blank is itself a fingerprint; decoy MUST be a complete small site	§3.3

#	Edge case	Required behavior	Cite
EC8	masque-h3 proxy SNI $\neq$ decoy cert name	forbidden — same name required or SNI distinguishes probe from client	§3.3
EC9	Hysteria2 < v2.9.2 (UDP ACL- bypass CVE- class)	pin $\geq$ app/v2.9.2 (2026-05-23); reject older	§5.3, [research- hysteria2 §1]
EC10	UDP_BLOCK + TLS_SNI_FILTER together (maximally hostile)	only shadowsocks/udp- over-tcp with tls_sni + AEAD front survive; if those fail $\rightarrow$ honest Down{reason} not a fake PASS	§2.1, §11.4.6

11. Error taxonomy extensions (regime-aware)

The base TransportError [01 §3.1 error.rs] is sufficient for the *carrier*; this document adds the **regime-classification** wrapping the orchestrator records (it does NOT change the wire taxonomy — EndpointBlocked is still the DPI verdict on the wire).

```
// helix-core/src/ladder.rs — orchestrator-side annotation, not a
// transport error
#[derive(Clone, Debug)]
pub struct RegimeVerdict {
    pub err: TransportErrorKind, // the carrier-level error
    // (mirrors §3.1 taxonomy)
    pub phase: FailPhase, // Dial | Handshake |
    // MidSession
    pub inferred: Regime, // bits this observation
    // contributes (may be empty)
    pub confidence: Confidence, // Hypothesis until a contrast
    // success confirms (§11.4.6)
    pub evidence_path: Option<String>, // §11.4.149 test_diary /
    // capture path
}
#[derive(Clone, Copy, Debug)] pub enum Confidence { Hypothesis,
    Confirmed }
```

EndpointBlocked remains the single wire-level "DPI/firewall dropped this" signal [01 §3.1]; the *regime* is an orchestrator inference layered on top, always Hypothesis until a differently-carried transport succeeds (then Confirmed, and the per-network memory short-circuits it next time) [§4.2, 01 §5.3 step 4].

12. What this document explicitly does NOT decide

- The **numeric** MASQUE throughput penalty — UNVERIFIED: pending the Phase-0 G2 benchmark [research-masque §5, 04_P0 §5.3]; Mullvad published none [research-hysteria2 §2].
- The **exact ClientHello mimicry** layer — an open design problem, seamed by SniStrategy (§6.4), specified for resolution in the security doc, not here [research-masque §2, §5].
- The **gateway edge language** (D5 Rust-vs-Go for MASQUE termination) — settled by the Phase-0 G4 benchmark; this document assumes Rust for the single-implementation guarantee and notes the Go (masque-go) conformance-suite-mitigated fallback [01 §11.2, research-masque §4].
- DAITA machine internals and the post-quantum PSK — referenced for seams only (§9), specified in the security doc [01 §9, research-daita].

End of nano-detail specification: Obfuscation & DPI/Censorship Landscape (Volume 2, Data Plane). Pair with the pass-1 overview [01 §3, §5.3] (the transports and the flat ladder this document makes regime-aware), the security doc (DAITA machines, ClientHello mimicry, PQ PSK), and Volume 8 testing docs (the §11.4.169 test-type harnesses behind §7). The Regime type (§1), the survives() matrix (§2), the probe AuthGate (§3), and the regime-aware ladder (§4) are the frozen contracts this document adds; their implementations may evolve.